ESPI



EXECUTIVE SUMMARY

CONVERGING CURRENTS

RECOMMENDATIONS TO ALIGN SPACE AND WATER POLICY AGENDAS IN EUROPE

CONVERGING CURRENTS

EXECUTIVE SUMMARY

This study identifies practical pathways for a stronger alignment between space and water policy priorities. It aims at informing ongoing and future policy processes, including the initiatives conducted in the framework of the European Water Resilience Strategy and the UN Water Action Agenda.

The results are derived from an analysis of national freshwater policy priorities across 32 European countries, as well as a compilation and review of over 250 primarily European-led downstream projects leveraging space data and infrastructure for freshwater-related applications, complemented by insights from stakeholder consultations.

National water policy priorities across European countries concentrate around mitigation of droughts and inland flooding, surface and groundwater pollution prevention, preservation of freshwater ecosystems, urban wastewater management and water infrastructure resilience, especially water supply management.

Satellite solutions already focus on surface water availability and hydrometeorological use cases such as floods and droughts, typically associated with high levels of insurance coverage, but still need to expand to more operational and localised applications for higher impact, addressing urban water management, water efficiency and sustainable water supply management. The integration of space applications with other technologies is proving essential but still untapped: only 20% of projects combine space with either AI, Internet of Things or Drones, addressing use cases such as irrigation management, green water and water losses.

HIGH CONTRIBUTION OF SPACE ACROSS THESE USE CASES:
Drought monitoring & management. Inland flood risk management, Green water availability, Surface water availability monitoring & management

At the intersection of the space and water policy communities, dialogue occurs across three main pathways: 1) top-down, initiated by national ministries or transboundary river basin authorities seeking technical support from international organisations, which connect requests with relevant geospatial expertise and networks, including international water conventions; 2) bottom-up, through satellite-based service coordinators engaging local water and disaster management actors, building capacity for using space-derived data and creating opportunities for service co-design, for instance hydrological predictions; 3) on the ground, with local actors supporting space applications by providing access to field sites, as well as in-situ information for satellite calibration and data validation.

Overall, a contrasted policy-project alignment is observed, with the level of prioritisation of water policy objectives being unevenly addressed by satellite solutions, in number and size.

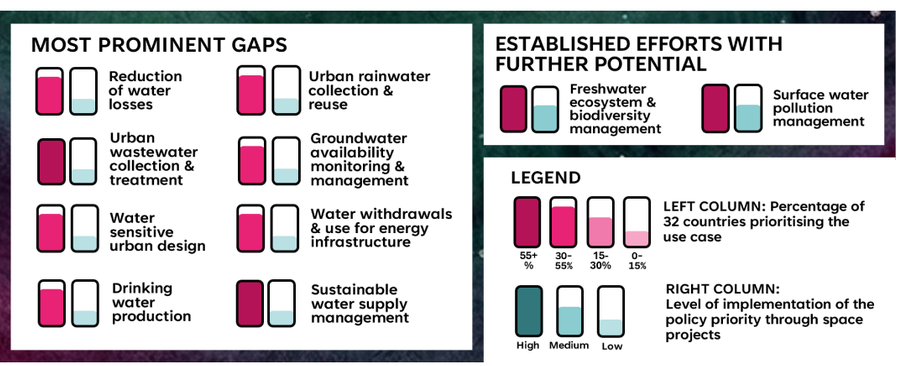
Key opportunities to expand space solutions emerge. In particular, the value of space data seems to be overlooked for ground-level operations in water supply and urban wastewater management, despite demonstrated potential (e.g., leak detection, water reservoir stability monitoring, detection of untreated sewage contaminating water resources). The focus of space solutions on floods, droughts and irrigation management instead suggests that the space sector remains largely focused on established use cases and “passive” (yet essential) monitoring, and has yet to transition towards more “proactive” solutions addressing less-explored operational needs.

Water diplomacy has emerged as a high-level policy concept reflected in a significant number of water policies, some specifically dedicated to it. However, it has yet to be explicitly recognised as an objective that space-based solutions can actively support, both within space policies and as the target of satellite solutions. In many transboundary contexts where negotiations over shared water resources management are hindered by either the absence of data, outdated data, or disagreements over the credibility of datasets, stakeholders consistently emphasised the current and potential value of Earth observation as a trusted, third-party information source, helping to reduce information asymmetries and build trust between riparian states.

PROMISING AVENUE:
Cooperation on Transboundary Water Basins

OPPORTUNITY ANALYSIS

Comparison of the level of prioritisation of water policy objectives and their implementation through space-enabled projects (full graph is in §2.4 of the report):



FINDINGS

Awareness of space applications within the water sector is high, but water-space policy alignment remains limited. While nearly 60% of analysed national water policies reference space technologies, mentions of water in space policies remain, in comparison, limited. Awareness gaps remain, as 20% of water policies mention only non-space technologies such as AI or drones. Overall, there is limited correlation between policy cross-reference and space projects implementation, which suggests that water and space policies evolve largely in parallel with limited coordination: water policies emphasise technological applications for water resource management (e.g., urban wastewater), while space policies tend to prioritise established use cases where the value of space is already demonstrated (e.g., droughts and inland flooding), rather than less-explored application areas (e.g., drinking water production, water supply management).

Water sector actors, public and private, are involved in only slightly over 25% of the consortia of analysed satellite-based projects. Public water actors (e.g., municipalities, public agencies), despite representing nearly half of primary end-users, are mostly absent from project consortia, creating a risk of technology providers disconnecting from implementation needs. Private water actors (e.g., water supply companies) remain also largely absent: when involved, they help connecting solutions with a broader spectrum of end-users, including hydropower producers, water supply companies and small-holder farmers.

Water management governance is decentralised, with implementation often occurring at basin and municipal levels. However, nearly 60% of analysed space projects are deployed at continent or global scale. While global, open-access platforms provide shared tools and datasets, local actors need tailored applications reflecting specific governance arrangements (e.g., river basin authorities) and water management challenges (e.g., wetland health monitoring). Limited funding for stakeholder mapping during project design reduces understanding of institutional landscape and implementation needs.

River basin organisations remain underrepresented in the consortium of space projects despite their strategic role in water management. Uniquely positioned to support national and transboundary water management, they bring valuable governance experience, GIS/HIS datasets, and long-term institutional knowledge. Integration of satellite data with these organisations’ ground observations would strengthen the work of river basin organisations beyond flood and drought forecasting, for freshwater ecosystem management and water quality; this requires capacity building, peer learning, and resource support to address differing basin challenges.

Recent water policies (2019-2025) increasingly emphasise water security, encompassing water availability, water infrastructure resilience and water diplomacy, yet these priorities remain insufficiently addressed by space projects. Despite its policy prominence, water infrastructure resilience features in a few projects, leaving opportunities in water supply management, water-sensitive urban design and drinking water production. Stakeholders noted that space data’s value is often overlooked in these activities traditionally viewed as ground-level operations, despite proven applications (e.g., reservoir monitoring, predictive maintenance of pipe networks). Similarly, while nearly 60% of space projects have an international impact reflecting the transboundary nature of water resources, water diplomacy remains largely absent from space policies and projects, despite being prioritised by nearly 30% of countries and the demonstrated value of space for transboundary cooperation.

Private companies offer untapped opportunities to finance and scale up water-related space solutions. Private water actors (e.g., private water suppliers, water management consultancies), involved in only 20% of projects analysed with little overlap with public water actors, provide access to end-users (e.g., insurers, large producers) who can help design and finance solutions. Water-dependent companies also have incentives to reduce their environmental footprint and support river basin preservation and restoration, as their degradation can be a consequence of their activities and a direct threat to their continuity. Private companies sometimes also sponsor publicly developed, open-access space-based platforms, stimulating downstream demand.

Space data delivers the greatest value when combined with in-situ data and embedded in broader governance frameworks. Stakeholders cautioned that adopting an exclusively technology-driven approach to freshwater management without recognising where ground data becomes necessary would risk “overselling” the potential of contribution of space technologies to freshwater applications, potentially leading to misdirected investments. Additionally, they underscored how such a technology-driven approach may trigger rebound effects: in agriculture, for instance, tools enabling more efficient water use often lead to the expansion of the cultivated area instead of reducing overall water consumption, especially when policy measures to reduce consumption remain limited.

RECOMMENDATIONS

1 POSITIONING SPACE AS A CORE COMPONENT OF WATER-MANAGEMENT SOLUTIONS

To encourage the development of solutions that directly address existing water policy challenges and improve the translation of policy priorities into space projects, there is a need to align cross-references between space and water policies around shared priority areas, where relevant. National space policies should explicitly recognise water as a strategic application area, while in countries promoting technology-enabled water solutions without reference to space, strengthening space considerations in water policies can help create entry points for the development of satellite-based solutions aligned by design with sectoral priorities.

2 CO-DESIGNING SPACE SOLUTIONS WITH WATER SECTOR ACTORS

Public and private water actors should be systematically involved as co-designers of water-related space projects rather than only consultees. Involving end-users throughout project development would strengthen trust, operational relevance and buy-in of solutions.

3 CONNECTING SPACE SOLUTIONS WITH WATER GOVERNANCE FRAMEWORKS THROUGH LOCAL ACTORS

Space solutions should be designed around the decentralised nature of water governance frameworks, informed by funded and mandatory stakeholder mapping. To enable this process, local and regional water actors should act as intermediaries to translate end-user needs into tailored solutions.

4 COORDINATING ACTION AT THE RIVER BASIN LEVEL

River basin organisations should be positioned as co-designers of space projects for water management, leveraging existing networks and peer-learning initiatives. Sustained dialogue with space providers should focus on needs identification and capacity building, moving from ad hoc collaboration towards shaping a shared agenda, to integrate space technologies into river basin management.

5 BRIDGING THE WATER-SECURITY-DIPLOMACY AWARENESS & OPERATIONAL GAP

The space sector should increasingly develop solutions addressing the operational needs of the water sector, particularly the security and resilience of water infrastructure. This should include stronger engagement at municipal level, where water supply is primarily managed, in anticipation of the growing prominence of water security in future policies. In parallel, the role of space technologies in diplomatic efforts and cooperation over shared water resources should be more explicitly recognised and translated into concrete objectives, both in water and space policies, and at implementation level.

6 LEVERAGING PRIVATE ACTORS AND FINANCE

Opportunities to co-finance water-related space solutions with private companies, and to increase the participation of private water actors in space projects consortia, should be actively explored. Such partnerships would help broaden the reach of space solutions and scale up applications addressing operational needs and preventive action (e.g., leak detection, water supply management).

7 RECOGNISING THE LIMITS OF SPACE TECHNOLOGIES AND MAXIMISING THEIR ADDED VALUE

Space-based services should complement, rather than substitute, in-situ data and local expertise. Integrating space and ground-based data will support adoption, while pairing investments with policy reforms on water allocation, pricing and abstraction limits will ensure that efficiency gains genuinely improve freshwater management.

FULL REPORT COMING SOON: WWW.ESPI.EU